

Systematic Options: The Wheel Strategy on NSE Derivatives

Research Report — Brindco Quant Researcher Take-Home Assignment

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Scope. This report reads the current outputs from outputs_real/, produced from downloaded NSE equity and derivatives bhavcopies through 2026-06-30. Verified split, bonus, rights, demerger, and symbol events are applied to the wheel and equal-weight universe. The action screen and limitations below define the remaining gaps.

1. Overview and verdict

The table below is generated directly from outputs_real/summary.csv. NIFTY 50 uses the official NSE Indices total-return series, which reinvests constituent dividends. The selected-universe benchmark includes recorded dividends and non-cash entitlements.

Series	End value	CAGR	Annual vol	Sharpe	Sortino	MDD	Calmar
Wheel portfolio	INR 47,405,473	14.21%	11.82%	0.66	0.57	-26.95%	0.53
NIFTY 50 total return	INR 42,283,722	12.22%	17.99%	0.38	0.35	-38.27%	0.32
Universe buy-and-hold	INR 66,785,977	20.40%	26.41%	0.61	0.55	-44.15%	0.46

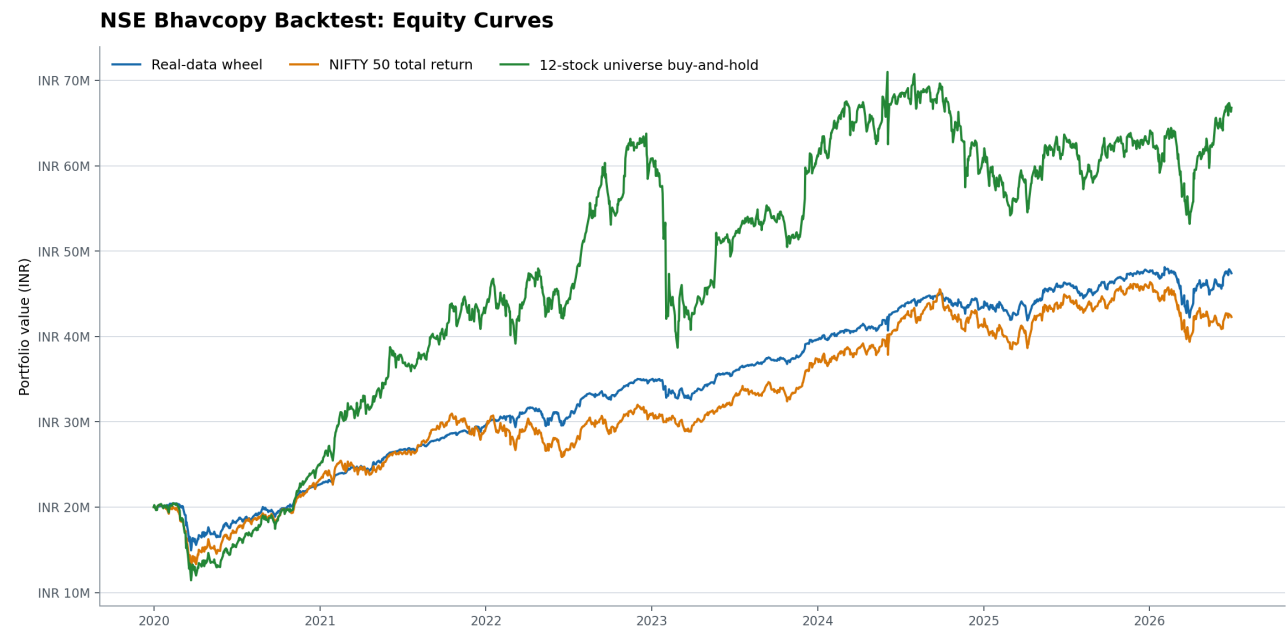


Figure 1. Portfolio equity curves from the real-data run.

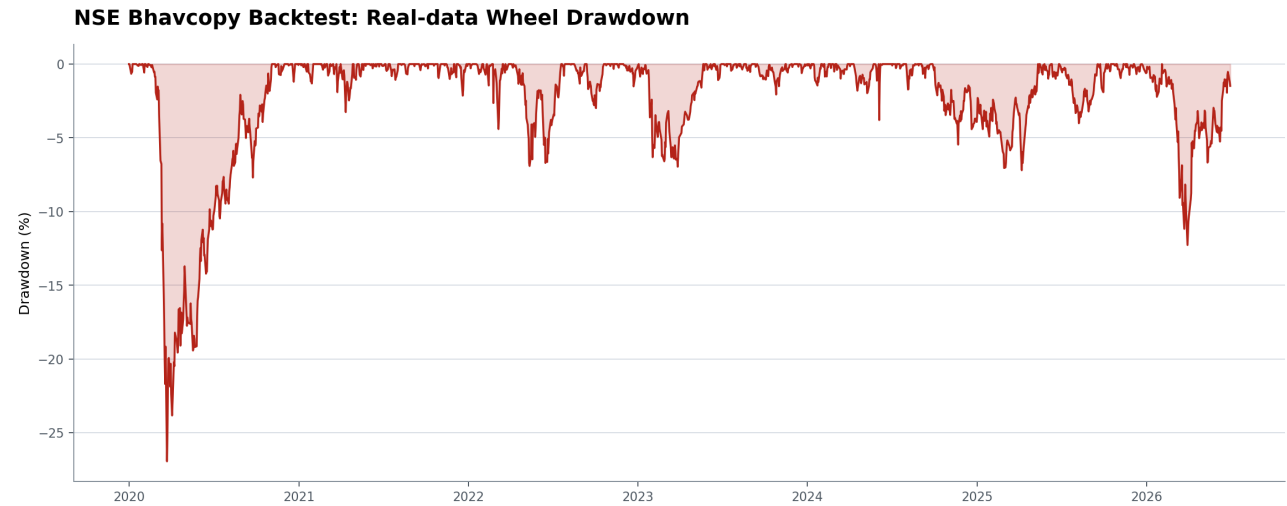


Figure 2. Drawdown calculated from the wheel equity curve.

2. Part A strategy specification and diagnostics

Capital and entry. INR 20 million is split equally across 12 fixed sleeves. Each sleeve waits until the second price date, uses the previous trading day's adjusted underlying close as its strike reference, and models sale at that day's observed option close. A contract needs positive premium, volume and open interest. The nearest listed monthly expiry must be at least seven calendar days away; size is rounded down to whole exchange lots. Unused cash earns an assumed 6.5% annual rate, accrued over 252 sessions.

Wheel rules. With no shares, sell the liquid put nearest 5% below the reference close and reserve the full strike obligation. An ITM put at expiry delivers shares at strike. With shares, sell the call nearest 5% above the reference close, never below the assigned net cost basis; if no qualifying quoted strike exists, sit out. An ITM call delivers the covered shares at strike and any residual shares are sold to reset to cash. Otherwise the expired leg is replaced in the next eligible session. There are no discretionary early profit takes, stops or rolls. Mandatory assignment has delivery costs but no execution slippage; modeled market sales do. A 20% option-margin proxy rises to 35% for ITM deliverables in the final seven calendar days, with pledged shares valued at 80% of close.

The engine recorded **1,681 trade-log events**, including 433 short-put entries, 100 assignments, 404 covered-call entries, and 89 call-aways. It inferred lots for 616 option entries, carried 23 stale marks, and recorded 11 expiry-date revisions.

Ticker	CSPs	Assign %	Call %	Avg days	Capture	Premium	Costs
RELIANCE	38	21.05%	17.07%	154.6	-4.71%	INR 1,664,906	INR 45,065
HDFCBANK	41	19.51%	19.44%	138.7	-10.81%	INR 1,091,155	INR 44,203
ICICIBANK	48	12.50%	17.24%	124.8	27.17%	INR 1,510,570	INR 31,652
TCS	32	31.25%	28.12%	84.9	12.81%	INR 1,544,796	INR 60,943
INFY	25	16.00%	6.98%	336.0	-18.40%	INR 863,190	INR 17,881
SBIN	40	25.00%	29.03%	118.3	-40.18%	INR 2,130,716	INR 57,101
AXISBANK	24	37.50%	16.33%	195.6	-67.36%	INR 1,772,132	INR 57,507
KOTAKBANK	33	39.39%	26.67%	105.4	-60.46%	INR 1,507,820	INR 67,799
LT	52	13.46%	28.57%	116.7	-4.22%	INR 1,467,562	INR 48,487
TATAMOTORS	26	26.92%	19.35%	157.5	15.07%	INR 1,933,964	INR 43,509
BAJFINANCE	49	26.53%	54.17%	67.9	-55.96%	INR 2,634,681	INR 93,869
ADANIENT	25	20.00%	18.18%	92.8	4.25%	INR 2,581,682	INR 30,944

CSPs counts short-put entries. Avg days covers completed delivery-to-call-away episodes. Capture is realized option premium less intrinsic settlement losses, divided by total premium sold; open contracts enter the denominator until expiry.

Per-name risk and return

Ticker	CAGR	Vol	Sharpe	Sortino	MDD	Peak	Trough	Recovery	Calmar
RELIANCE	13.39%	15.92%	0.48	0.41	-23.73%	2020-03-03	2020-03-23	2020-04-22	0.56
HDFCBANK	12.14%	13.90%	0.44	0.37	-20.49%	2020-03-03	2020-03-24	2020-07-06	0.59
ICICIBANK	14.82%	16.81%	0.54	0.41	-30.92%	2020-03-03	2020-03-23	2020-11-10	0.48
TCS	8.18%	15.13%	0.18	0.15	-44.00%	2024-12-13	2026-06-30	not yet	0.19
INFY	4.57%	15.84%	-0.04	-0.04	-28.71%	2025-01-10	2026-06-30	not yet	0.16
SBIN	16.66%	14.23%	0.72	0.56	-28.44%	2020-02-26	2020-05-22	2021-02-01	0.59
AXISBANK	17.21%	26.00%	0.51	0.44	-51.34%	2020-03-03	2020-03-24	2021-01-07	0.34
KOTAKBANK	14.40%	16.35%	0.52	0.45	-21.95%	2020-03-05	2020-03-23	2020-10-27	0.66
LT	17.10%	17.23%	0.64	0.50	-37.08%	2020-01-31	2020-03-24	2020-12-14	0.46
TATAMOTORS	9.91%	22.21%	0.25	0.22	-50.85%	2020-02-05	2020-04-03	2020-11-18	0.19
BAJFINANCE	20.31%	18.31%	0.77	0.62	-35.14%	2020-03-03	2020-05-26	2020-11-12	0.58
ADANIENT	16.81%	28.01%	0.48	0.41	-53.51%	2023-01-09	2023-02-27	2024-05-23	0.31

Each sleeve begins with INR 20 million / 12; 'not yet' means unrecovered by the end date. Exact figures are in outputs_real/per_name_summary.csv.

Reconciled P&L decomposition

P&L component	Portfolio INR
Option premiums	INR 20,703,175
Stock delivery + mark	INR -3,544,033
Dividends	INR 1,063,008
Rights value	INR 73,736
Spun-off stock	INR 1,268,382
Cash interest	INR 8,440,166
All costs	INR -598,961
Open option liability	INR 0
Ending equity (incl. INR 20m start)	INR 47,405,473

Realized stock delivery P&L was INR 4,616,767; it is a subset of the stock line, which also marks remaining shares to the final close. Adding it again would double-count returns.

3. Data and methodology

The downloader checked weekdays from 2020-01-01 through 2026-06-30. It found 1,604 price dates, 1,603 option dates, and 4,891,323 option rows. 91 weekdays had all three NSE reports missing; one date had prices but no options report.

Universe selection

The original model defined a fixed 12-name F&O shortlist spanning energy, banks, IT, industrials, autos, and NBFCs. TCS represented the lower-volatility sleeve; TATAMOTORS and ADANIENT supplied cyclical and idiosyncratic stress. We retained this qualitative choice and tested option participation on 2020-01-01 before the first strategy entry. Eligibility required at least 250 option contracts traded, 1 million summed share open interest, and 15 rows with both volume and open interest; all 12 passed. Exact readings are in data/universe_selection.csv.

Assignment gap. This checks liquidity only within an already chosen shortlist. It does not satisfy the brief's requested objective selection from the full F&O candidate pool, and the fixed names have survivorship bias. A compliant new universe needs an ex-ante all-market screen and a rebuilt backtest; the present performance should be read with that limitation.

Coverage check	Value
Price dates	1,604
Option dates	1,603
Option rows	4,891,323
Duplicate contract keys	0
Price cells missing	0
Missing bid/ask fields	4,891,323 / 4,891,323
Zero volume fields	3,143,833

Execution uses traded option closes and prior-day stock closes for strike choice. Lot sizes come from NSE fields or checked legacy inference. Dividends and corporate events are applied on ex-dates; short-option sale STT and share-delivery STT are charged. Puts are fully cash-secured and calls share-covered. The 20%/35% SPAN-plus-delivery proxy is checked daily against cash and haircut share collateral. Full assumptions are machine-readable in outputs_real/assumptions.json.

For splits and bonuses, held share and open-option quantities scale by the exchange factor while strike, premium and cost basis scale inversely. The rights event uses the NSE strike and lot adjustment and credits modeled rights value; demergers create a separate spun-off claim, provisionally marked then sold at its first NSE listing close. Recorded cash dividends are credited to prior-close holdings on the ex-date. Expiry resets and symbol changes are recorded in the action log.

Corporate-action screen for all twelve names

Ticker	Recorded event(s)	Largest adjusted 1-day drop
RELIANCE	2020-05-13 RIGHTS; 2023-07-19 EXPIRY_RESET; 2023-07-20 SPIN_OFF; 2024-10-28 BONUS	13.15% (2020-03-23)
HDFCBANK	2025-08-26 BONUS	12.61% (2020-03-23)
ICICIBANK	No large action found	17.85% (2020-03-23)
TCS	No large action found	9.41% (2020-03-12)
INFY	No large action found	10.04% (2020-03-23)
SBIN	No large action found	14.4% (2024-06-04)
AXISBANK	No large action found	27.91% (2020-03-23)
KOTAKBANK	2026-01-14 SPLIT	13.0% (2020-03-23)
LT	No large action found	16.27% (2020-03-23)
TATAMOTORS	2025-10-13 EXPIRY_RESET; 2025-10-14 SPIN_OFF; 2025-10-24 SYMBOL_CHANGE	14.36% (2020-03-23)
BAJFINANCE	2025-06-16 SPLIT_BONUS	23.23% (2020-03-23)
ADANIENT	2025-11-17 RIGHTS	28.2% (2023-02-01)

The screen compares adjacent raw closes after applying recorded entitlements. It can identify large discontinuities; it cannot establish that every small corporate distribution is captured. Cash dividends are extracted separately from the NSE action register.

Interpretation limits. Bid and ask quotes were absent, so entry and mark prices use traded close and settlement fields. The 104 dated dividend records in data/dividends.csv are credited to shares held before each ex-date session. Rights are monetized at NSE theoretical values; spun-off shares use the special pre-open implied value until listing, then NSE closes. Margin is a proxy, not a historical exchange SPAN replay; see outputs_real/daily_positions.csv. Performance is exploratory, not a production claim.

4. Sensitivity and stress narrative

Scenario	Strike OTM	Slippage	CAGR	Vol	Sharpe	MDD
base	5.00%	25 bps	14.21%	11.82%	0.66	-26.95%
strike_3pct	3.00%	25 bps	13.40%	12.35%	0.57	-26.61%
strike_8pct	8.00%	25 bps	12.14%	11.05%	0.53	-26.23%
slippage_10bps	5.00%	10 bps	14.23%	11.82%	0.66	-26.94%
slippage_50bps	5.00%	50 bps	14.18%	11.83%	0.65	-26.95%

The strike choice matters: moving from 5% to 3% OTM lowers Sharpe from 0.66 to 0.58; moving to 8% lowers it to 0.53. The 10-50 bps slippage range changes CAGR by only about 0.05 percentage points, but that narrow modeled range cannot validate fills when every bid/ask field is missing.

The wheel's worst drawdown was -26.95%, from 2020-02-13 to 2020-03-23, with recovery on 2020-11-04. Cash-secured short puts were already open at the peak. At the 27 February expiry, the rules forced two assignments, in TATAMOTORS and LT; both then sold covered calls. Stock exposure rose from INR 982,350 to INR 1,928,676. The five largest sleeve losses and endpoint positions are below.

Sleeve	Peak position	Trough position	Stock move	Equity change
AXISBANK	0 shares; short put 690	0 shares; short put 700	-58.73%	INR -852,707
TATAMOTORS	0 shares; short put 175	8600 shares; short call 170	-60.94%	INR -840,778
LT	0 shares; short put 1300	1125 shares; short call 1300	-43.97%	INR -607,468
ICICIBANK	0 shares; short put 505	0 shares; short put 490	-47.50%	INR -510,323
BAJFINANCE	0 shares; short put 4100	0 shares; short put 4500	-52.74%	INR -495,844

AXISBANK lost INR -852,707 without holding shares at either endpoint: its open short put was marked against a -58.73% underlying move. TATAMOTORS and LT lost through delivered shares after the two assignments. The trade log shows 12 new option sales and INR 4,952 of logged event costs during this episode. Daily option liabilities, cash and shares are in outputs_real/daily_positions.csv.

5. Part B: synthetic index wheel design

NIFTY and BANKNIFTY options are cash-settled, so a stock-style physical wheel cannot be copied directly. A concrete synthetic construction is: sell a cash-secured OTM NIFTY put; if it finishes ITM, buy one NIFTY futures lot and carry the futures position as the synthetic delivered index; while the futures position is held, sell an OTM call; if the call finishes ITM, close the futures lot at the call strike and return to the cash-secured put state. If either option expires out of the money, retain the premium and sell the next monthly contract. The implementation must mark futures basis daily, charge futures roll costs, enforce option and futures margin offsets, and round every position to the current lot size. Weekly expiries can raise premium turnover and gamma risk; monthly expiries align more closely with the stock-wheel cycle. The index wheel removes single-name gap risk but concentrates market beta and basis risk. This design is included as required; it is not presented as an implemented result.

A precise state transition would use the expiry-day index settlement S and short-put strike K . If S is below K , the put settles for a cash debit of $(K-S)$ times the current lot size. The system then buys one matching near-month futures lot, subject to next-session execution and basis slippage, and keeps enough free cash for daily variation margin. That combined exposure behaves approximately like index ownership from K , but cannot deliver the index basket itself. When long futures are held, calls are written against the same lot count. If S exceeds a short-call strike, the cash option debit is offset by the futures gain; the futures are closed and the cash-secured put state resumes. Call strikes must also respect the synthetic position's net entry level, so a large index decline can leave calls far out of the money and premiums small.

For a test, choose one index and one contract tenor ex ante; use dated NSE options and futures settlements, actual historical lot sizes and expiry calendars, and the same close-to-next-open execution convention for every transition. Maintain a separate cash ledger for premiums, option cash settlement, futures daily mark-to-market, financing, transaction charges, STT, and slippage. Reserve worst-case put cash obligation and upfront option/futures margins; after an ITM put, test whether the account can fund the new futures margin plus a stress buffer before entering. If it cannot, record a missed transition rather than silently assuming leverage. Roll the futures before expiry, explicitly paying the old/new basis spread and fees. A weekly schedule needs more frequent re-hedging, more event risk near expiry, and higher turnover; the monthly schedule is the closer analogue to the stock-wheel experiment.

Report the index-wheel equity alongside NIFTY 50 total return, not just the price index, and compare both variants under common strike and slippage sensitivities. A futures overlay introduces basis, roll, and margin-call risk even though it removes stock-specific dividends and corporate actions. NIFTY 50 and BANKNIFTY should be evaluated separately because contract size and volatility change cash requirements. The index contracts are cash-settled and futures are marked to market daily under the NSE settlement mechanism. This section defines an implementable study; no index-wheel performance is claimed here.

6. Limitations, sources and reproducibility

Daily equity and F&O bhavcopies come from NSE reports; cache paths and missing dates appear in `data/source_manifest.json`, while archive URL patterns are in `main/data_pipeline/fetch_nse_data.py`. Event links are in `data/corporate_actions.csv`. Benchmark: NSE Indices historical TRI. Tax: NSE STT rates and payer. Margin: NSE margin FAQ. The 20%/35% coefficients are proxies, not daily exchange SPAN calculations. Unlisted strikes and stale option marks are possible; bid/ask fields are absent. Previous-day stock prices avoid strike-selection look-ahead, but filling at the same day's observed option close remains optimistic. Fixed 2026-observable names introduce survivorship bias.

Deployability verdict. The wheel returned 14.21% annualized versus 12.22% for NIFTY 50 total return, but the selected stock buy-and-hold earned 20.40%. Interest on idle or secured cash contributed INR 8,440,166 of the wheel's INR 27,405,473 gain. The evidence is not sufficient for a production deployment claim until the universe is selected objectively and executions/margins are tested with more realistic data.

Run `python3 -m main.data_pipeline.fetch_nifty_tr`, then `python3 -m main.backtest.run_real_backtest`, `python3 -m main.backtest.plot_real_results`, and `python3 -m main.report.build_report`. Corporate-data utilities are grouped under `main/data_pipeline/`; tests are isolated under `tests/`. Install `requirements.txt` first.